

BIGSCREEN BEYOND 2

VR/AR Technology Review By Alba Samsami

PC VR Headset

107g Ultra-Light

Micro-OLED

Eye Tracking

What Is Bigscreen Beyond 2?

Bigscreen Beyond 2 is a tethered PC VR headset released in 2025 by Bigscreen Inc., a company originally founded in 2014 that began as a Social VR movie-watching app.

Its core purpose is to deliver the highest visual quality in the smallest, lightest possible form factor. Unlike standalone headsets, it acts as a pure display peripheral connected to a powerful gaming PC.

The headset is the gateway to both the SteamVR ecosystem (thousands of PC VR games and apps) and Bigscreen's own social cinema platform.

Founded

2014 — Bigscreen Inc.

Announced

March 2025 — GDC

Headset Type

Tethered PC VR (SteamVR)

First-Year Sales

Beat Beyond 1 in 10 days

Who Is It For?

PC VR Enthusiasts

Users with SteamVR setups who want best visuals without a heavy headset

Movie & Media Fans

People using Bigscreen app to watch films in shared virtual cinema rooms

Sim Racers & Pilots

Simulation enthusiasts with wheels, HOTAS, or cockpit setups

VRChat / Social VR

Avatar-based social worlds; eye tracking enables lip sync & expressions

VR Productivity

Users streaming PC desktops as floating virtual multi-monitor workspaces

NOT For Beginners

Requires PC, base stations & controllers. Not plug-and-play or for kids.

Device & Specifications

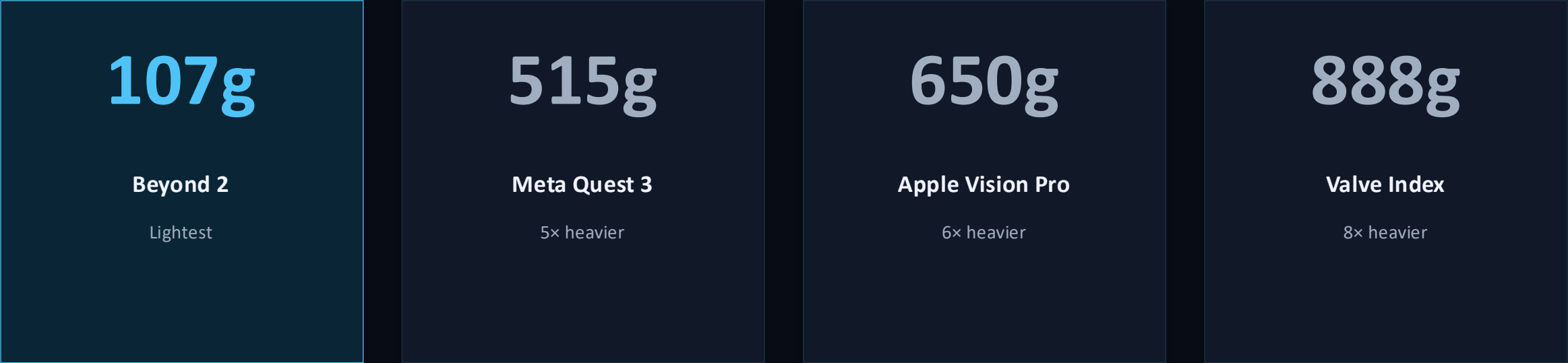


© Bigscreen Inc. — Beyond 2 with Holomount

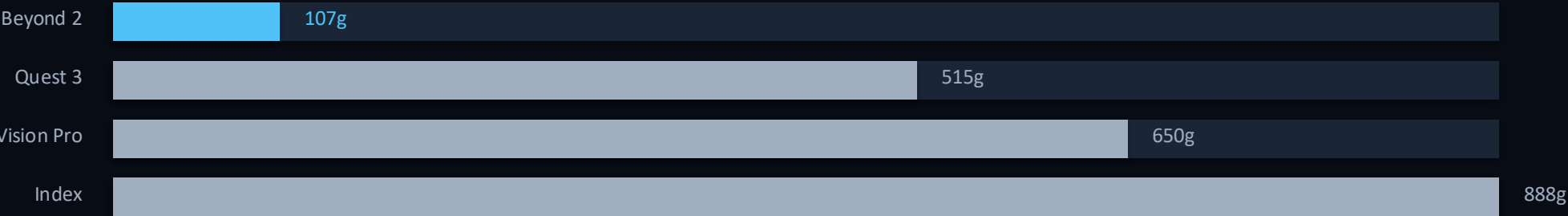
Display	2× 1" Micro-OLED panels
Resolution / Eye	2560 × 2560 pixels
Lenses	Pancake Optics
Field of View	116° diagonal
Refresh Rate	75 Hz / 90 Hz
Weight	107g (headset only)
Tracking	Base Stations
IPD Range	48 – 75 mm (adjustable)
Price (Beyond 2)	\$1,019 USD

ULTRA-LIGHT DESIGN

Weight Comparison



Relative Weight (Beyond 2 = 107g baseline)



Visual Experience



© Bigscreen — Micro-OLED lens pair

Display Tech

Dual 1-inch Micro-OLED panels, perfect blacks, vibrant colors, deep contrast. 2560×2560/eye. Sharper than Quest 3 LCD.

Pancake Optics

New lenses provide edge-to-edge clarity, large sweet spot, and massively reduced glare vs. original Beyond fresnel lenses.

Rendered Resolution

At 100% SteamVR supersampling: 3560×3560 per eye, fantastic detail without requiring a top-of-the-line GPU.

Brightness Tradeoff ⚠

Pancake lenses reduce light by ~50%. OLED compensates in dark scenes but bright/outdoor scenes appear dimmer.

What Can You Do?

Social Cinema (Bigscreen App)

Watch movies on massive virtual screens with up to 16 friends in shared virtual rooms. Paramount, Lionsgate, and other studio partnerships.

SteamVR Gaming Library

Access thousands of SteamVR games, Half-Life: Alyx, Beat Saber, Flight Simulator VR, iRacing, and more.

Virtual Desktop

Stream your PC desktop into VR as multiple floating virtual monitors in any environment.

Eye Tracking (Beyond 2e)

AI-powered eye tracking (90Hz, 400×400/eye). Adds only 1.05g. Foveated rendering, gaze interaction, VRChat eye expressions.

Room-Scale 6DoF Tracking

Full 6-degree-of-freedom positional tracking via SteamVR base stations , walk freely in your physical space.

Full-Body Tracking

Compatible with HTC Vive Trackers and Tundra Trackers for full-body avatar tracking in social VR.

Bigscreen Social Cinema App



© Bigscreen Inc. — Virtual movie theater environment

The Bigscreen App: Original Product, Still Central

Bigscreen's Social VR cinema app turns your headset into a massive private or shared movie theater.

- ▶ Watch on a 200+ inch virtual screen
- ▶ Invite up to 16 friends simultaneously
- ▶ Themed virtual rooms — theaters, living rooms, outdoors
- ▶ Paramount, Lionsgate & other studio partnerships
- ▶ Voice chat and avatar facial expressions
- ▶ Available free — no subscription required

How Users Interact

INPUT METHODS

SteamVR Controllers

Valve Index or HTC Vive 6DoF tracked controllers for gaming. Highly accurate, near-zero latency tracking. Not included, purchased separately.

Sim Peripherals

Steering wheels, flight sticks, and HOTAS setups. Controllers entirely optional for cockpit-style simulations.

Bigscreen Laser Pointer

Virtual laser pointer UI for navigating cinema app menus. Simple and effective for non-gaming use.

What It Can't Do

No Standalone Mode

Requires a tethered VR-ready PC. Cannot be used away from a computer.

No Passthrough / MR

No cameras on the headset. Completely isolated from the real world, no mixed reality.

High Total Cost

\$1,019 headset + \$149 base stations + \$299 controllers + PC = \$2,000+ total.

No Bundled Controllers

No controllers in the box. Users must source SteamVR-compatible controllers separately.

Brightness Reduction

Pancake lenses cut ~50% light. Bright outdoor scenes appear noticeably dimmer.

Complex Setup Required

Mount 2+ base stations, run 5m cable, configure SteamVR. Not plug-and-play.

Design & Implementation Choices

✓ Tethering as a Feature

Offloading processing to a PC enables micro-OLED displays, top-tier tracking, and zero battery anxiety, at the cost of mobility.

✓ Pancake Lens Choice

Pancake optics dramatically reduce optical stack depth, enabling the headset's tiny form factor. The brightness tradeoff is mitigated by OLED's self-emissive pixels.

✓ SteamVR Ecosystem

Leveraging the existing SteamVR ecosystem means thousands of compatible games and apps on Day 1, no proprietary app store to populate from scratch.

What Should Come Next?

Wireless Streaming

A Wi-Fi 7 or UWB wireless solution would remove the cable, the single biggest friction point, without compromising latency.

Higher-Brightness OLED

As OLED tech matures, higher-nit panels would offset pancake lens light loss, making daylight scenes more vivid.

Foveated Rendering

Full eye-tracked foveated rendering could cut GPU load 40–60%, enabling higher resolutions on mid-range hardware.

Hand Tracking

Adding cameras for inside-out hand tracking would reduce reliance on separate controllers for casual cinema use.

120 Hz+ Refresh Rate

A jump to 120Hz+ would improve smoothness in fast-paced games, making the experience feel more fluid.

Expand Bigscreen App

More studio partnerships, live event streaming, and mixed-reality desktop integration to strengthen the social cinema value proposition.



BIGSCREEN BEYOND 2

The Lightest Premium PC VR Headset

107g weight

2560×2560 per eye

116° FOV

Micro-OLED

SteamVR ecosystem

bigscreenvr.com